Group Project Code

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1. Set up

Packages

```
# general use
library(tidyverse)
library(janitor)

# file organization
library(here)
```

Data

```
# weather data from NOAA weather station
weather <- read_csv(here("data", "NOAA-weather-data.csv"))

# water quality survey metadata
metadata <- read_csv(here("data", "NCOS_YSI_Water_Quality_Monitoring_0.csv"))

# water quality parameter measurements
water_parameters <- read_csv(here("data", "YSI_Data_Begin_1.csv"))
```

Code

```r
weather_clean <- weather |>
  # made column names all lowercase
  clean_names() |>
  # puts date in proper format and makes sure its understood as date (not as chr)
  mutate(date = mdy(date)) |>
  # keeping only these 4 columns of interest
  select(date, prcp, tmax, tmin) |>
  # creating new columns for month and year
  # extracting month from the date (needs to be in the standard date format for this functio
  mutate(month = month(date),
         year = year(date)) |>
  mutate(wy = case_when(
    # when month os october or later, assign a value of year +1
    # in the water ywar for next year
    month >= 10 ~ year + 1,
    # keeping all months < 10 the same (sept or earlier), assigning value of existing year
    .default = year
    ))
```

```r
metadata_clean <- metadata |>
  # col names easier to work with (all lowercase w/ _)
  clean_names() |>
  # "!" is the "not-operator", columns we DONT want to keep, excluding
  select(!c(object_id, creator, editor)) |>
  mutate(monitoring_date = mdy_hms(monitoring_date)) |>
  rename(monitoring_datetime = monitoring_date) |>
  mutate(month = month(monitoring_datetime),
         year = year(monitoring_datetime)) |>
  # assigning water year
  mutate(wy = case_when(
    month >= 10 ~ year +1,
    .default = year )) |>
  # creatig new column for site full name
  mutate(site_full = case_when(
    site_name == "east_channel" ~ "East Channel",
    site_name == "phelps_bridge" ~ "Phelps Bridge",
    site_name == "venoco_bridge" ~ "Venoco Bridge",
    site_name == "other" ~ "Other"
  )) |>
  # filter out Other observations
  filter(site_full != "Other")
```

```r
# creating a new object from water parameters
water_parameters_clean <- water_parameters |>
  # cleaning column names
  clean_names() |>
  # "!" not operator, excluding unneeded columns
  select(!c(creator, editor)) |>
  # renaming depth code to be elevation code to better represent methodology
  rename(elevation_code = depth_code) |>
  ## Everything below came from visualizing the data first and defining what to fix/add
  # fixing a mistake in the data
  mutate(elevation_code = case_when(
    # anytime theres a 10 in elevation code, replace it with a 1
    elevation_code == 10 ~ 1,
    # everything else, keep the same
    .default = elevation_code)) |>
  mutate(
    # turning column into a factor
    elevation_code = as_factor(elevation_code),
    # defining the correct order for the levels of the factor
    elevation_code = fct_relevel(elevation_code,
                                 # write in the order you want
                                 "0", "1", "2", "3", "4", "5", "6"))
```

2. Joining

water quality = metadata + water parameters

```r
# full joining
water_quality <- full_join(
  # left data frame
  x = metadata_clean,
  # right data frame
  y = water_parameters_clean,
  # name the columns in each df that you want to join by (naming the shared column)
  by = c("global_id" = "parent_global_id")) |>
  select(
    # information about the observation
    global_id, monitoring_datetime, time_of_wq_measurement, monitor_name, weather, site_name
    # water information
```

```
    water_surface_elevation, flow, notes,
    #measurement information
    elevation_code, do_mg_l, do_percent_sat, conductivity_u_s_cm, salinity_ppt, temperature_c
  ) |>
  # taking the date out of this column, dont want the time
  mutate(date = date(monitoring_datetime)) |>
  # moving the new date column to a better location within the df
  relocate(date, .after = monitoring_datetime) |>
  # dont need this column anymore
  select(!monitoring_datetime)
```

water quality weather = water quality + weather

```
# left join
water_quality_weather <- left_join(
  # left data frame
  x = water_quality,
  # right data frame
  y = weather_clean,
  # joining these dfs by their date column
  by = "date"
)
```

3. Wrangling

Dealing with Outliers in Salinity

```
# creating new obj
salinity_outlier_ids <- water_quality_weather |>
  # filtering to only include outlier values
  filter(salinity_ppt > 1410) |>
  # extracting the global id column as a vector usign pull
  pull(global_id)
```

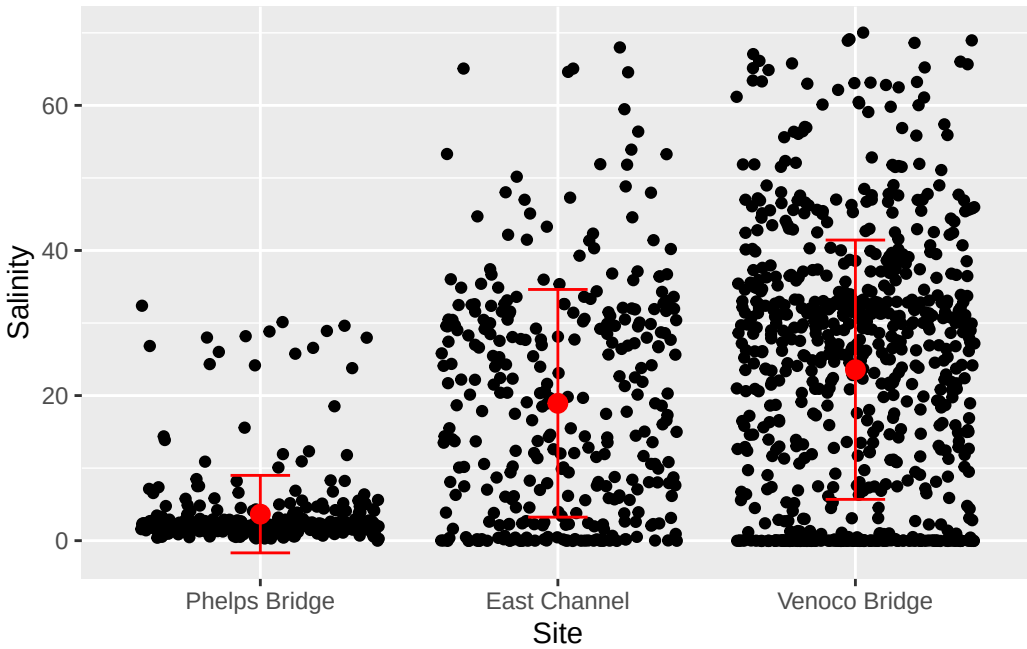
Filtering Salinity

```
salinity_filtered <- water_quality_weather |>
  filter(elevation_code %in% c("0", "1", "2", "3")) |>
  # filtering the column global_id to exclude the values that match those in `salinity_outlie
  filter(!global_id %in% salinity_outlier_ids) |>
  # setting the order of the levels (from lowest tp highest avg salinity value)
  # helps the order of the visualizations
  mutate(site_full = fct_relevel(site_full,
                                 "Phelps Bridge", "East Channel", "Venoco Bridge")) |>
  filter(!site_full == "NA")
```

4. Summarizing

5. Comparing groups Viz

```
ggplot(data = salinity_filtered,
       mapping = aes(x = site_full,
                     y = salinity_ppt)) +
  geom_jitter() +
  stat_summary(fun = mean, geom = "point", size = 3, color = "red") +
  stat_summary(fun.data = mean_sdl, fun.args = list(mult = 1), geom = "errorbar", width = 0.
  labs(x = "Site",
       y = "Salinity")
```



6. Figure Visualizing

Figure 1

```
ggplot(data = salinity_filtered,
       mapping = aes(x = date,
                     y = prcp)) +
  geom_point() +
  geom_line()
```

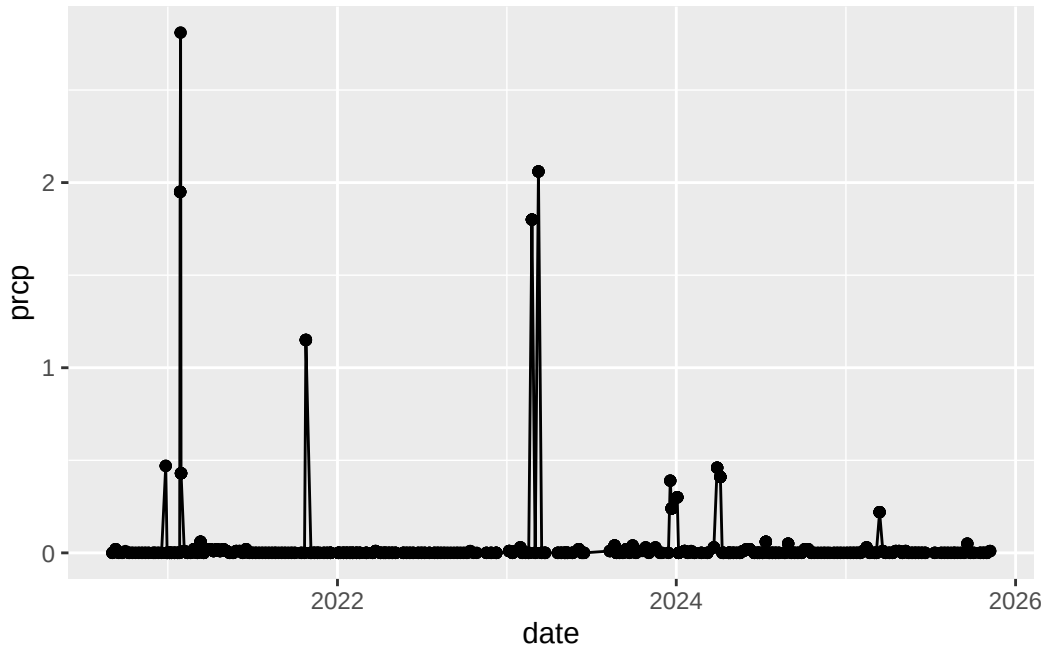


Figure 2.

```
ggplot(data = salinity_filtered,
       mapping = aes(x = date,
                     y = water_surface_elevation)) +
  geom_point() +
  geom_line()
```

```
Warning: Removed 706 rows containing missing values or values outside the scale range
(`geom_point()`).

Warning: Removed 2 rows containing missing values or values outside the scale range
(`geom_line()`).
```

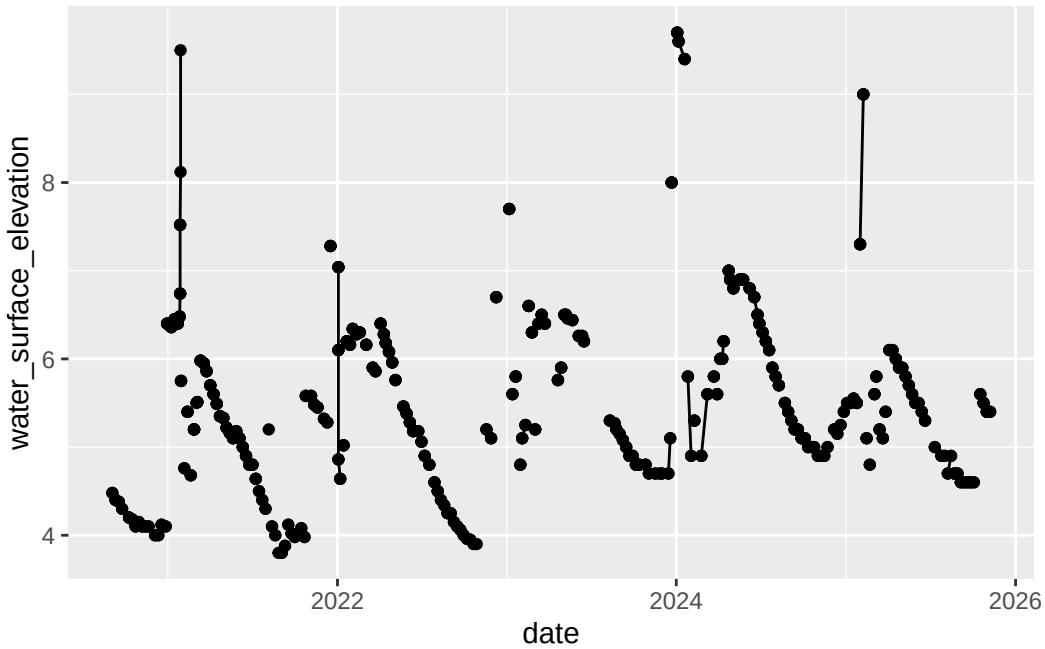


Figure 3.

```
ggplot(data = salinity_filtered,
       mapping = aes(x = salinity_ppt,
                     y = elevation_code)) +
  geom_col()
```

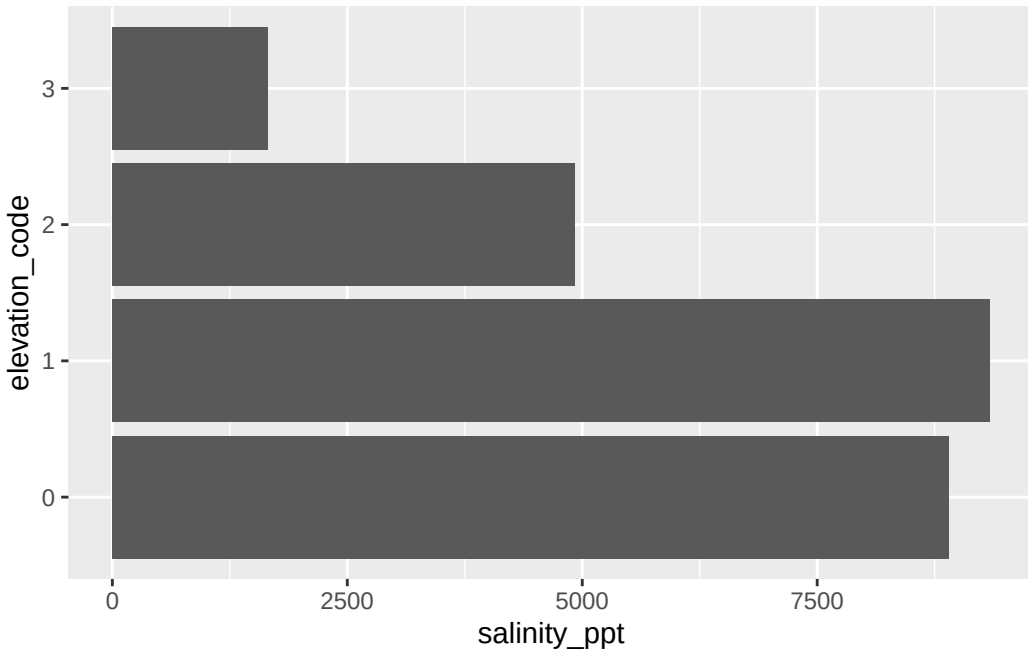
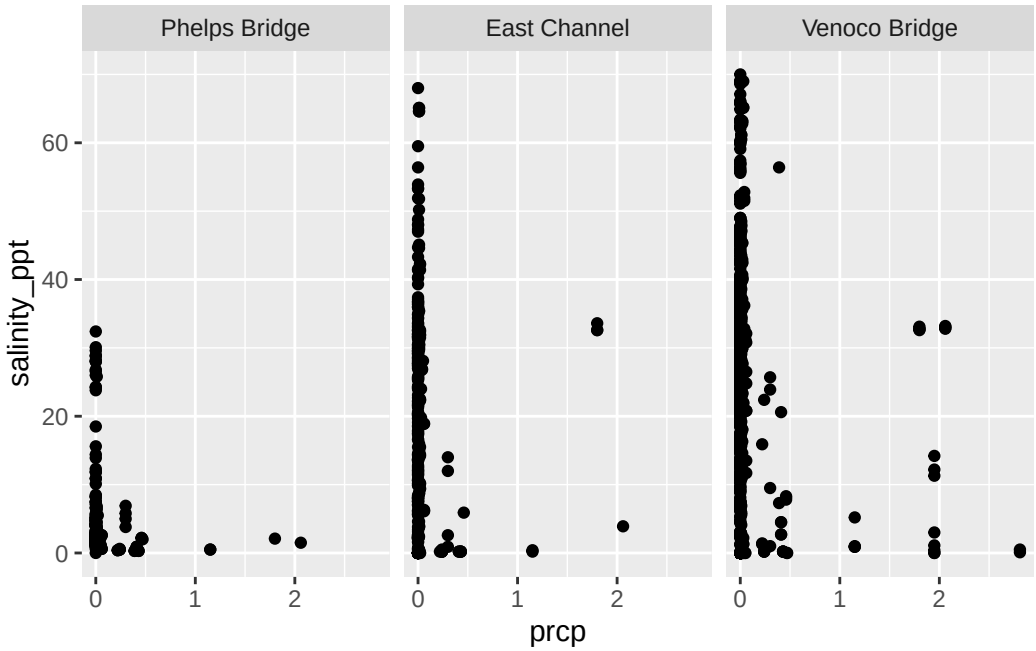


Figure 4.

```
ggplot(data = salinity_filtered,
       mapping = aes(x = prcp,
                     y = salinity_ppt)) +
  geom_point() +
  facet_wrap(~ site_full)
```


Phelps Bridge
East Channel
Venoco Bridge
salinity_ppt
60
40
20
0
0
1
2
prcp